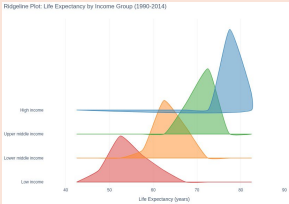
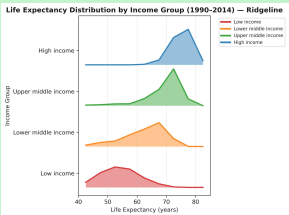
# DV-Evol Bad Case ploty.js：Visual mapping failure

**Question: Generate a ridgeline plot comparing life expectancy (1990–2014) across income groups. Use fixed 5-year bins from 40 to 85, normalize to percent frequency, and stack the ridges from "Low income" to "High income" using designated HEX colors.**

**Gemini-3-pro-preview:**




**Ground Truth:**




**Plot code:**

```
for g in groups:
    d = result_df[result_df['Group'] == g].sort_values('BinMid')
    off = y_offsets[g]
    px = [d['BinMid'].iloc[0]] + d['BinMid'].tolist() +
[d['BinMid'].iloc[-1]]
    py = [off] + (d['Percent'] + off).tolist() + [off]

    fig.add_trace(go.Scatter(
        x=px, y=py, fill='toself', mode='lines',
        line=dict(color=colors[g], width=1.5, shape='spline'),
        fillcolor=f"rgba{tuple(int(colors[g][i:i+2], 16) for i in (1, 3,
5)) + (0.5,)}"
    ))

fig.update_layout(
    xaxis_range=[35, 90],
    yaxis=dict(tickvals=list(y_offsets.values()), ticktext=groups,
range=[-5, 170]),
    plot_bgcolor="white"
)
```

**Error Analysis:** 1. Data Integrity

✅1.1 Trend/Pattern: 1/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

❌1.2 Completeness: 0/1 Critical data series or specific years are missing from the visualization.

❌1.3 Label Accuracy: 0/1 Axis tick values and labels do not precisely match the original data points.

❌1.4 Data Consistency: 0/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

❌2.5 Element Spacing: 0/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

❌3.1 Labels: 0/1 Legend and title positioning are misaligned compared to the standard layout.

✅3.2 Readability: 1/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

❌3.3 Axis Alignment and Spacing: 0/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

❌3.4 Element Alignment: 0/1 Graphical components fail to align precisely with their respective axis ticks.

❌3.5 Proportionality and Aspect Ratio: 0/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

❌3.6 Overall Balance: 0/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

## Score: 4/16